

STUDY GUIDE FOR THE FINAL EXAM

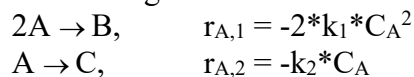
Instructions: The final exam will be **closed book/closed notes** covering **all topics** from the entire semester. The test will last 2.5 hours.

While the exam is closed notes, you can bring two “torpedo” sheets from the previous 2 tests and a new torpedo for this final exam with any information of your choosing (8.5” x 11”, you can use both sides). The torpedoes must be created by you. At the end of the exam, staple your torpedo sheets to the back of the exam.

Review the two previous study guides as well as your exams. In addition to these, you should be prepared to address the following:

- **Deriving differential equations for transient reactive systems.** Apply *molar balances* to derive the differential equations for a closed or open system in which chemical reactions take place. Identify initial conditions. One reaction may take place or multiple reactions may take place concurrently. Be able to use the given **order of chemical reaction** information to calculate the generation (G) and consumption (C) terms in the general material balance.

Example: A continuously stirred tank reactor (CSTR) is filled with 10 L of a 1 M solution of species A. Liquid continuously enters and exits the tank at a constant flow rates of $\dot{v}_0$ and $\dot{v}$, respectively. Starting at $t=0$, species A decomposes into species B and species C according to the following reactions and reaction rates:



In the above equations, $r_{A,i}$ is rate of generation of A in reaction i. Assume that all species are in the liquid state, and the density of the liquid ρ remains constant. Use initial conditions to generate the set of differential equations describing how the concentration of each species changes with time. Show your work.

- **Numerical solution of differential equations.** Apply numerical methods (*e.g.*, Euler, Predictor-Corrector, and 2nd order or 4th order Runge-Kutta methods) discussed in class to solve individual or **coupled** differential equations numerically.

Example: Write the equations for C_{i+1} , k_1 , and k_2 if you use the 2nd order Runge-Kutta method to numerically solve the following coupled differential equations:

- $dC_A/dt = \dots$
- $dC_B/dt = \dots$

Note: *always using corresponding k for updating the matched variables* (*e.g.*, use C_A 's k_1 and k_2 to update C_A , and C_B 's k_1 and k_2 to update C_B)

- **Numerical integration and differentiation.** Given a set of data, perform numerical integration using the rectangular rule, trapezoidal rule, Romberg integration, and Simpson's 1/3 and 3/8 rule. For the same data, perform numerical differentiation using the forward, backward, and central methods. Graphically illustrate each rule or method.

Example: Apply the Romberg rule to integrate across the following data set representing the concentration of species A over time:

t (min)	C _A (mM)
0	100
1	70
2	35
3	50
4	65

- **Excel VBA.** We will not ask you to write a VBA code without using a computer. However, we may give you a simple code and ask to interpret what it does, fill in blank lines, identify bugs and provide corrections. Know what is a macro and VBA, and how to record a macro. Be able to perform the following in VBA:
 - Define a new variable with correct data format (single, double, integer, string, etc.)
 - Read variable values from textbox or spreadsheet
 - Output results to textbox or spreadsheet
 - If/Else and If/Elseif statement
 - Select Case statement
 - For loop
 - Do loop
 - Use GoTo command to exit For loop
 - Use Exit Do command to exit Do loop
- **Matlab.** Basic syntax for creating .m files (script and function files). Define a new function or multiple functions at the same time. Call a function in a script. Create and manipulate matrices in Matlab. Use Matlab to solve series of linear equations. Use the fzero and fsolve functions to solve individual or multiple nonlinear equations. Use the ode45 function to numerically solve individual and coupled ordinary differential equations.

Example: What is the objective of entering `fsolve(@example1, [0 0])` on the command line, where `example1.m` has the following code:

```
function y = example1(x)
y = [x(1) - x(2) - 2; x(1) + x(2) - 4];
end
```

- **Do not forget to review the earlier content and examples covered in class** (e.g., steady state CSTR with multiple reactions, transient CSTR with multiple reactions, tanks of different shapes being filled, and so on, going back through the semester).
- **Do not forget to review all homework assignments and midterm tests.**